Parameter recovery: acuity selection on the tested grid

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Parameter recovery: acuity selection on the tested grid

Parameter recovery probes whether the executed observation model contains enough information to distinguish sensor acuity under the study design. We sweep acuity values 0.60, 0.70, 0.80, 0.90:

Parameter recovery (part 2)

for each true acuity the model generates 200 synthetic observations per trial across 960 independent trials, fits acuity by marginal-likelihood grid search, and compares the recovered value with ground truth.

Parameter recovery (part 3)

The configured seed initializes the base RNG stream; it is not the independent analysis unit.

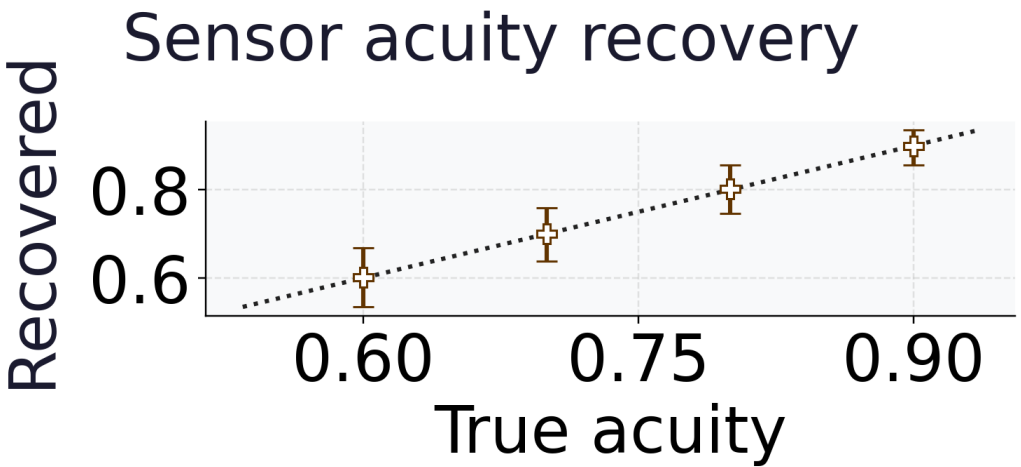
Parameter recovery (part 4)

Across the sweep the mean absolute recovery error is 0.0232 and the coefficient of determination of mean-recovered versus true acuity is $R^2 = 0.9999$. Within this finite grid and observation budget, recovered acuity tracks the identity line with the reported error (fig. 34).

Parameter recovery (part 5)

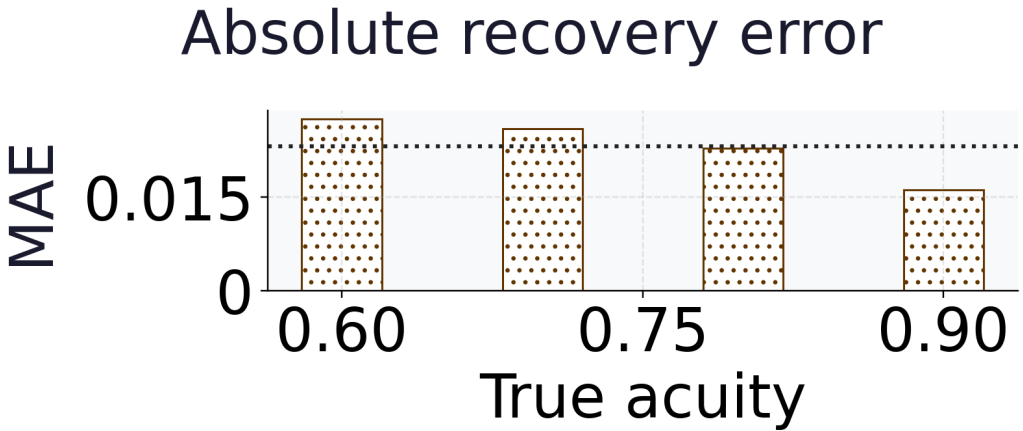
This is evidence of practical acuity recoverability for the executed design, not a proof of global or structural identifiability and not an acuity-by-colony-size study.

Parameter recovery (part 6)



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Parameter recovery (part 7)



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Parameter recovery (part 8)

Recovery whiskers: 95% empirical percentile intervals over independent simulation trials within each acuity. Observations are nested within trial.

Parameter recovery (part 9)

independent trials/acuity = 960
observations/trial = 200
MAE = 0.0232

Parameter recovery (part 10)

$$R^2 = 0.9999$$

Parameter recovery (part 11)

Simulation-based recovery over the declared acuity grid. Identity reference is not an additional estimate; no uncertainty interval is inferred for MAE bars.